

Data Science

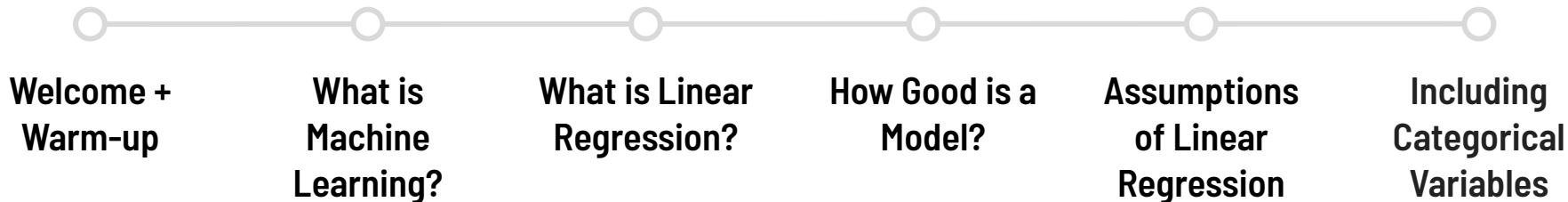
Linear Regression

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Describe the principles of linear regression.

2

Interpret the coefficients of a linear regression model.

3

State the assumptions of a linear regression model.

4

Fit, evaluate, and generate predictions from a linear regression model in **sklearn**.

5

Include categorical variables in a linear regression model.

What is Machine Learning?

An overview of the key concepts

AI or not?



AI or not?



Zoopla estimate

Want to know what your home could be worth right now? Our Zoopla estimate will give you a guide price, based on analysis of the richest sources of property data in the UK.

[Get a Zoopla estimate](#)

AI or not?

How much can I borrow?

Get a rough idea of how much you could borrow for a residential mortgage based on your personal circumstances.

How many applicants? [?]

☒ 1 ☐ 2 ☐ 3 ☐ 4

My income [?]

£ Yearly

Regular spending

The amount you spend to repay credit and store cards, catalogue purchases, loans, overdrafts, maintenance and your pension. You don't need to tell us about general household spending, such as groceries, travel and utility bills.

£ Monthly

Reason for mortgage

Please select

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG
PILE OF LINEAR ALGEBRA, THEN COLLECT
THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

JUST STIR THE PILE UNTIL
THEY START LOOKING RIGHT.



What is Machine Learning?

Machine learning is the process of letting your **machine** use data to **learn** the relationship between some **predictors** (x-variables) and some **response**(s) (y-variables).

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This is a fancy way of saying **statistical modeling**.

Ok then, so what is statistical modeling?

Statistical modeling is the process of combining data with **statistical theory** to **model** the real-world relationship between **predictors** (x-variables) and some **response**(s) (y-variables).

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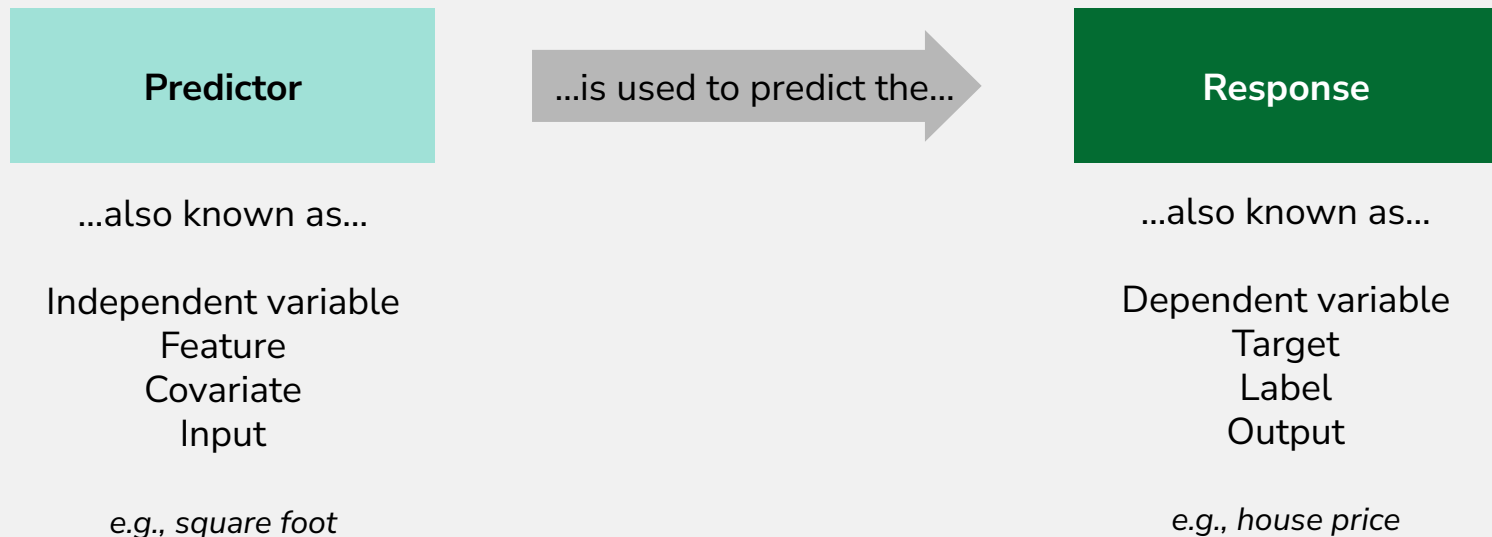
Statistical modeling is the process of combining data with **statistical theory** to **model** the real-world relationship between **predictors** (x-variables) and some **response(s)** (y-variables).

This is a more down-to-earth way of saying **machine learning**.

For example, you might want to predict:

- The sale price of a house based on features like size, location, and number of rooms, or
- Whether a customer will cancel their subscription in the next month based on their recent activity patterns.

Because Machine Learning has evolved from a number of fields, many things have many names...



Terminology

For each of these examples, which is the **independent variable**, and which is the **dependent variable(s)**?

- Modeling the effect of coffee consumption on a student's final grade
- Predicting a company's share price using their profits
- Predicting a company's gender pay gap using the proportion of women on the company's board and the company sector

There are different types of variables

Categorical data:

- a. Ordinal: natural order exists (ratings, education levels)
- b. Nominal: no natural order (colours, cities, brands)

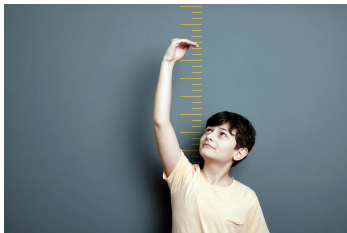
Numerical data:

- a. Discrete: count data (number of bedrooms, clicks)
- b. Continuous: measured data (height, temperature, price)

Terminology

Are each of these variables **categorical** or **numerical**? Are they **ordinal**, **nominal**, **discrete** or **continuous**?

Height



Eye color



Weight



Month



Two* Kinds of ML

In essence, all machine learning models fall into one of two categories:

- **Supervised learning** - Given X, can we predict Y?
- **Unsupervised learning** - What does X look like, *really*? There is no Y.

** There are more. Kinda. The big third category is **reinforcement learning**, which is a field still in the process of being invented.*

Two Kinds of Supervised Learning

Supervised learning models fall into two different buckets:

Regression - this is when our y-variable is numeric.

- *“Given the past values of the stock price of Apple, what will tomorrow’s closing price be?”*
- *“Given the annual precipitation, average temperature, and soil pH, what will this year’s harvest yield be?”*
- *“Given the square footage, number of bedrooms, number of bathrooms, and quality of school district, what will the price of this home be?”*

Two Kinds of Supervised Learning

Supervised learning models fall into two different buckets:

Classification - this is when our y-variable is a category. If it's a 0/1 yes/no kind of variable, we often call it **binary classification**. Otherwise, **multiclass classification**.

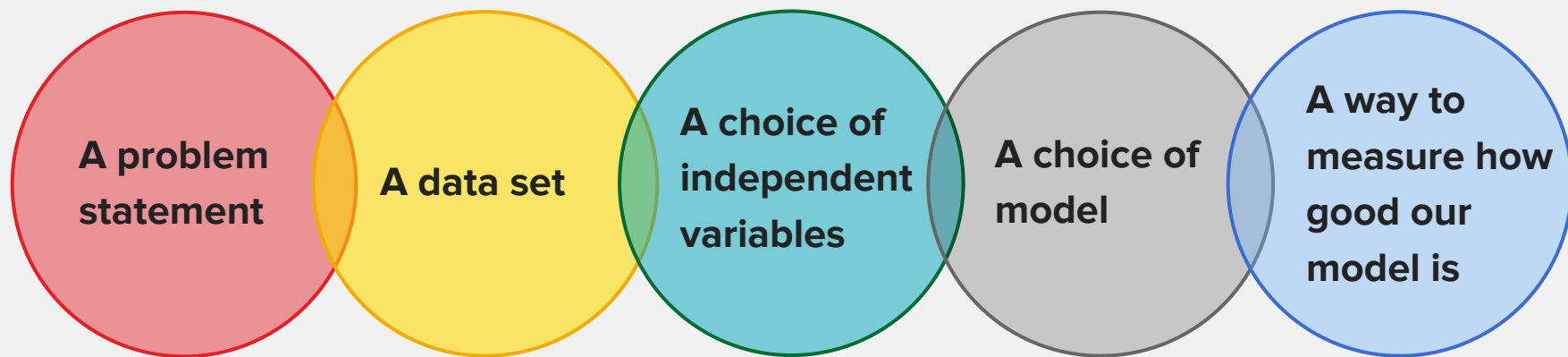
- *“Given this person’s demographic information, how many tabs they have open, and where they live, will they make a purchase on my site?”*
- *“Given radar readouts, past weather, and almanac data, will it rain tomorrow?”*
- *“Given how many hours you study, how many hours you sleep, and your course load, will you pass the final exam?”*

Classification or regression?

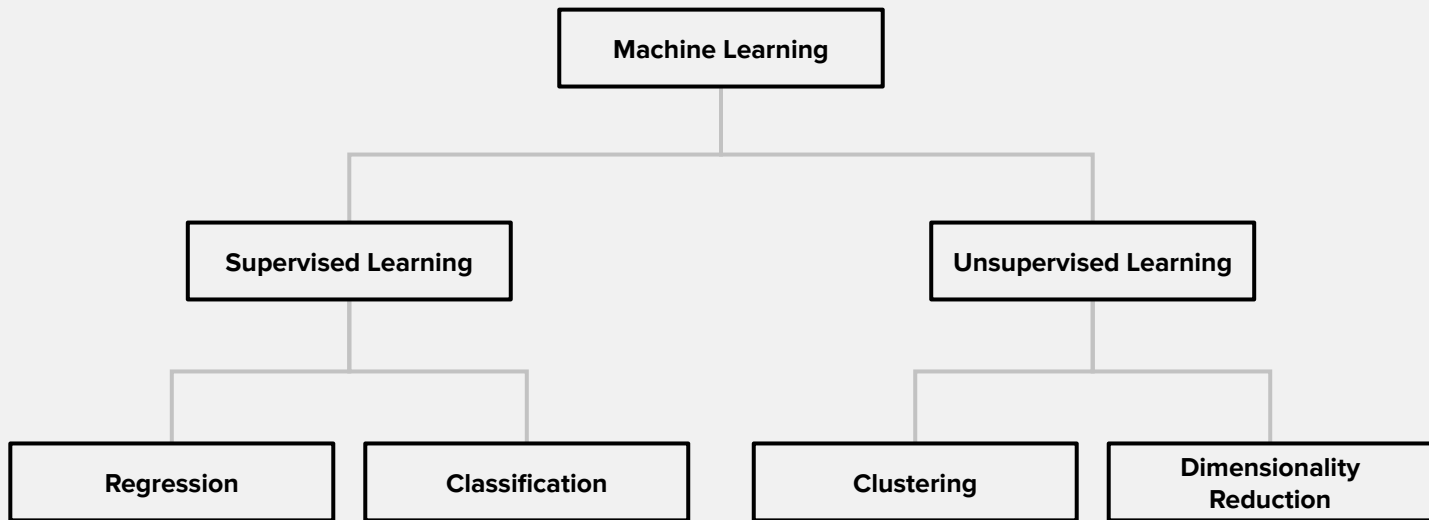
For each of these machine learning problems, decide whether it's a **classification** or **regression** task, and identify what the **features(s)** and **response** are:

Problem	
1	Predicting daily rainfall (mm) using temperature, air humidity and wind speed
2	Predicting a house sale price using its location and number of bedrooms
3	Predicting a hair colour using their eye colour and age
4	Forecasting tech sales on Black Friday
5	Predicting power plant failures

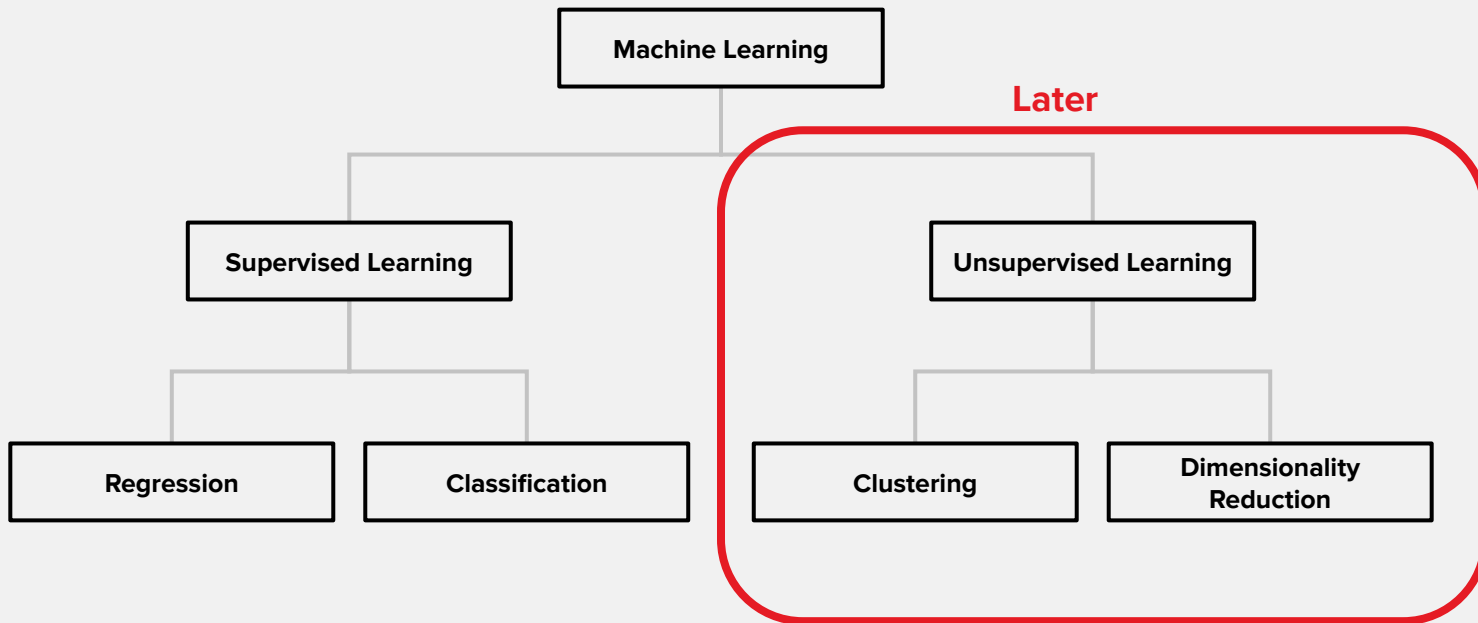
Building Supervised Models: What Do We Need?



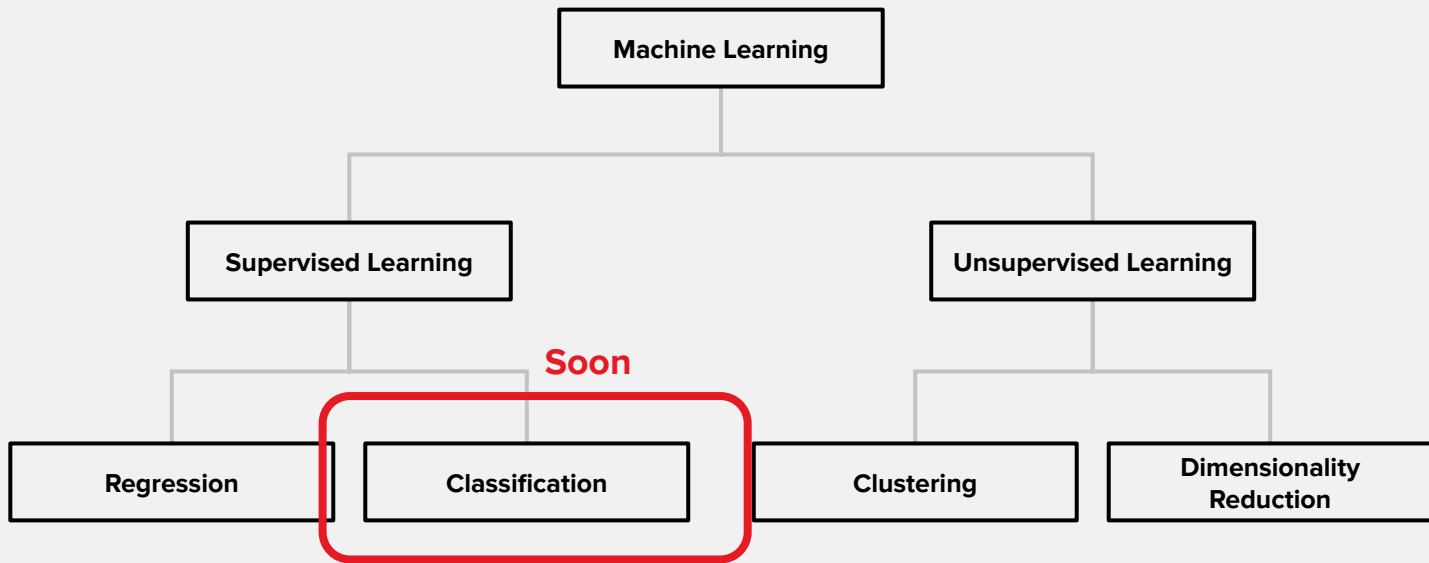
Roadmap of ML



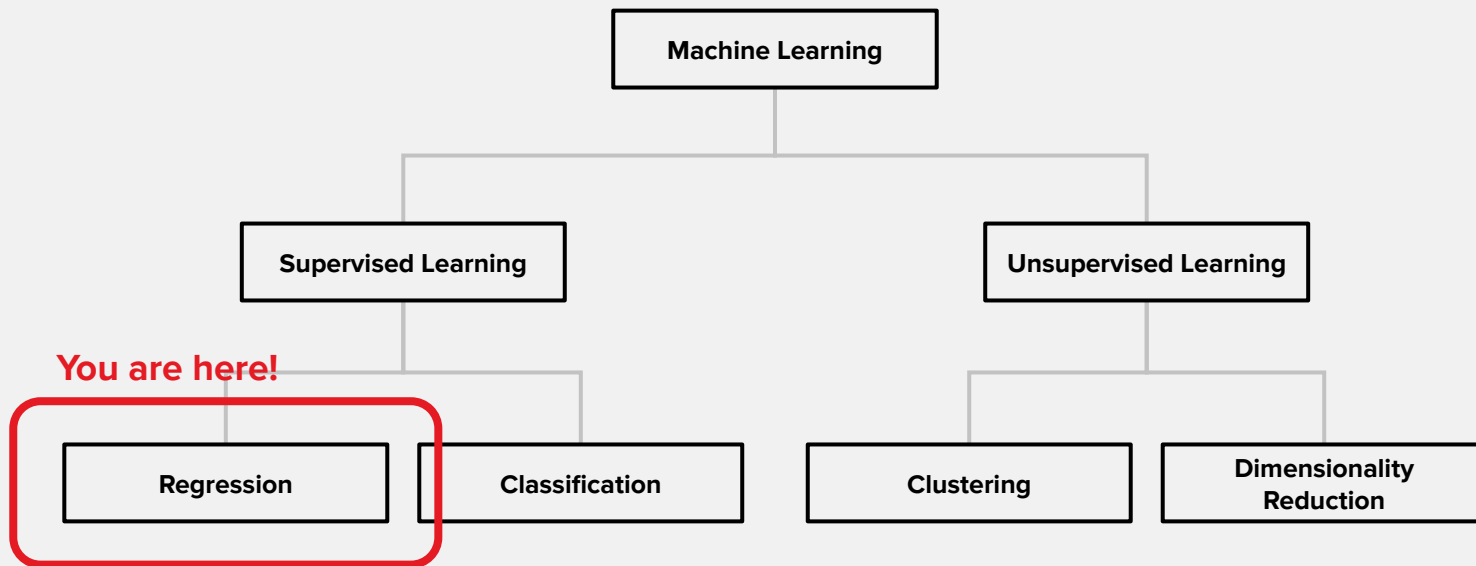
Roadmap of ML



Roadmap of ML



Roadmap of ML



Supervised Learning Transparency

Profitable Results!

Data



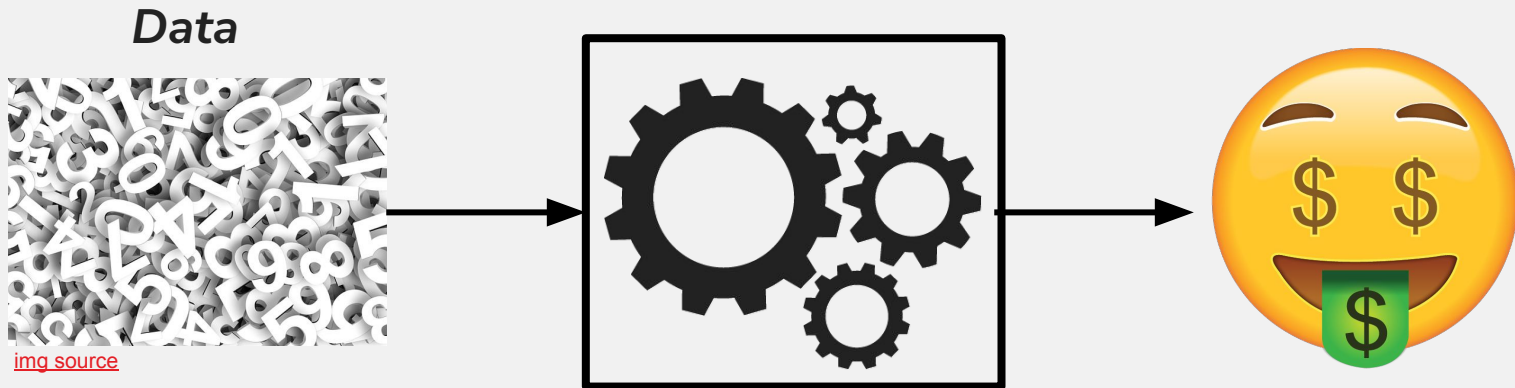
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Supervised Learning Transparency

Profitable Results!

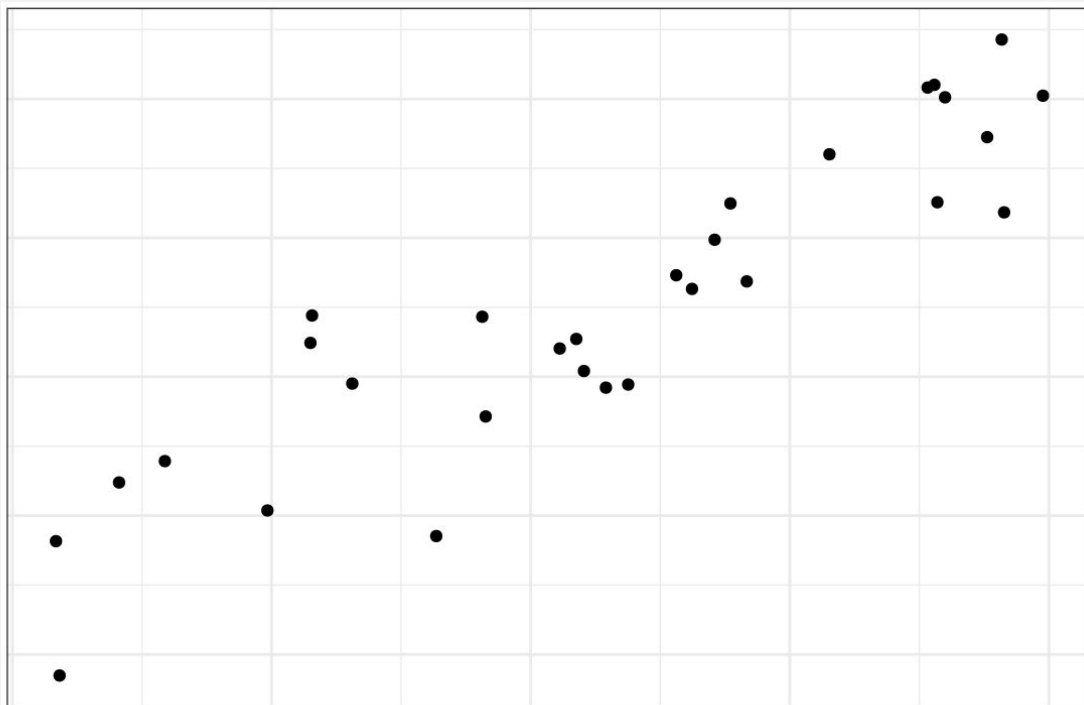


What is Linear Regression?

Supervised, white-box, regression

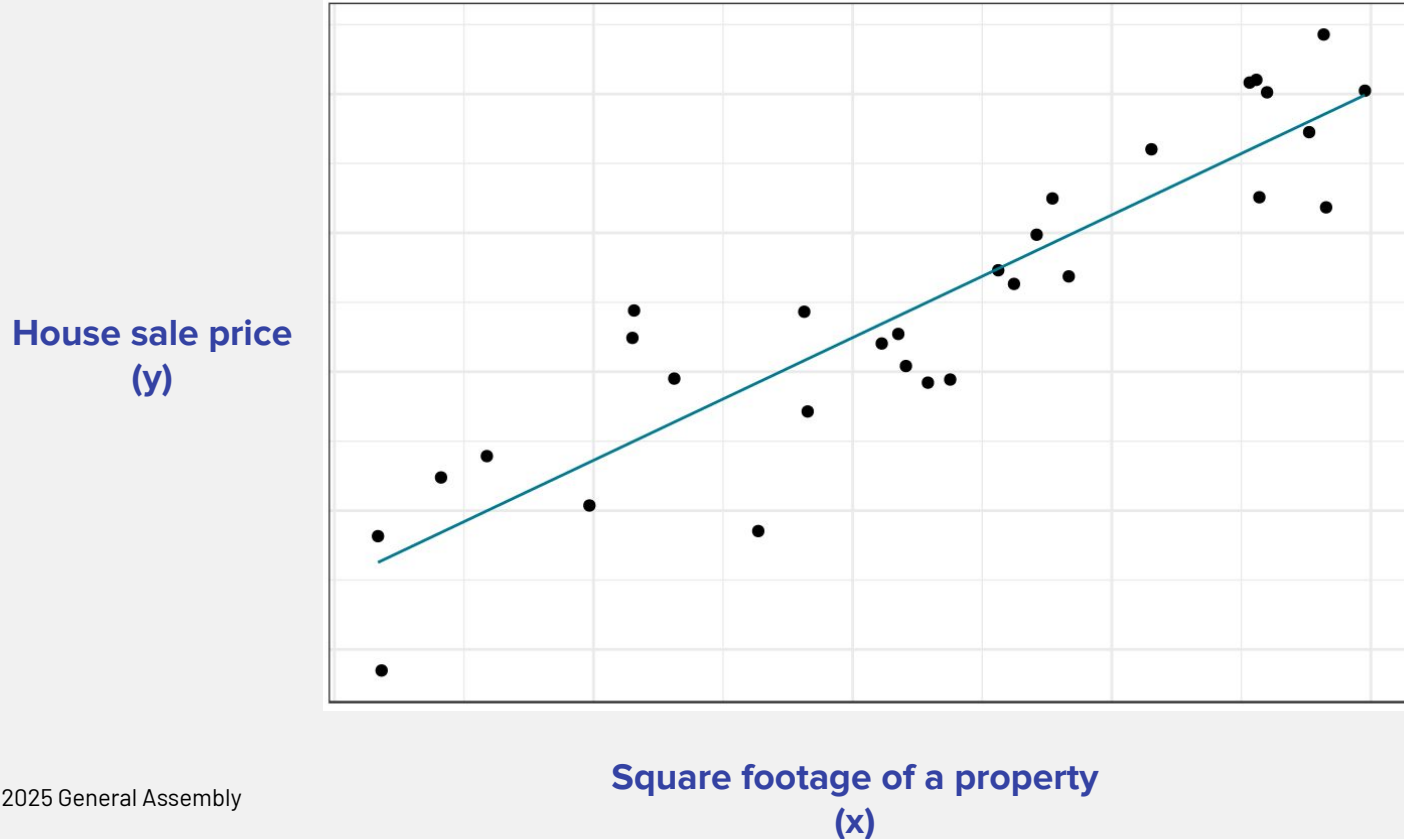
Let's look at an example of two variables we want to model

House sale price

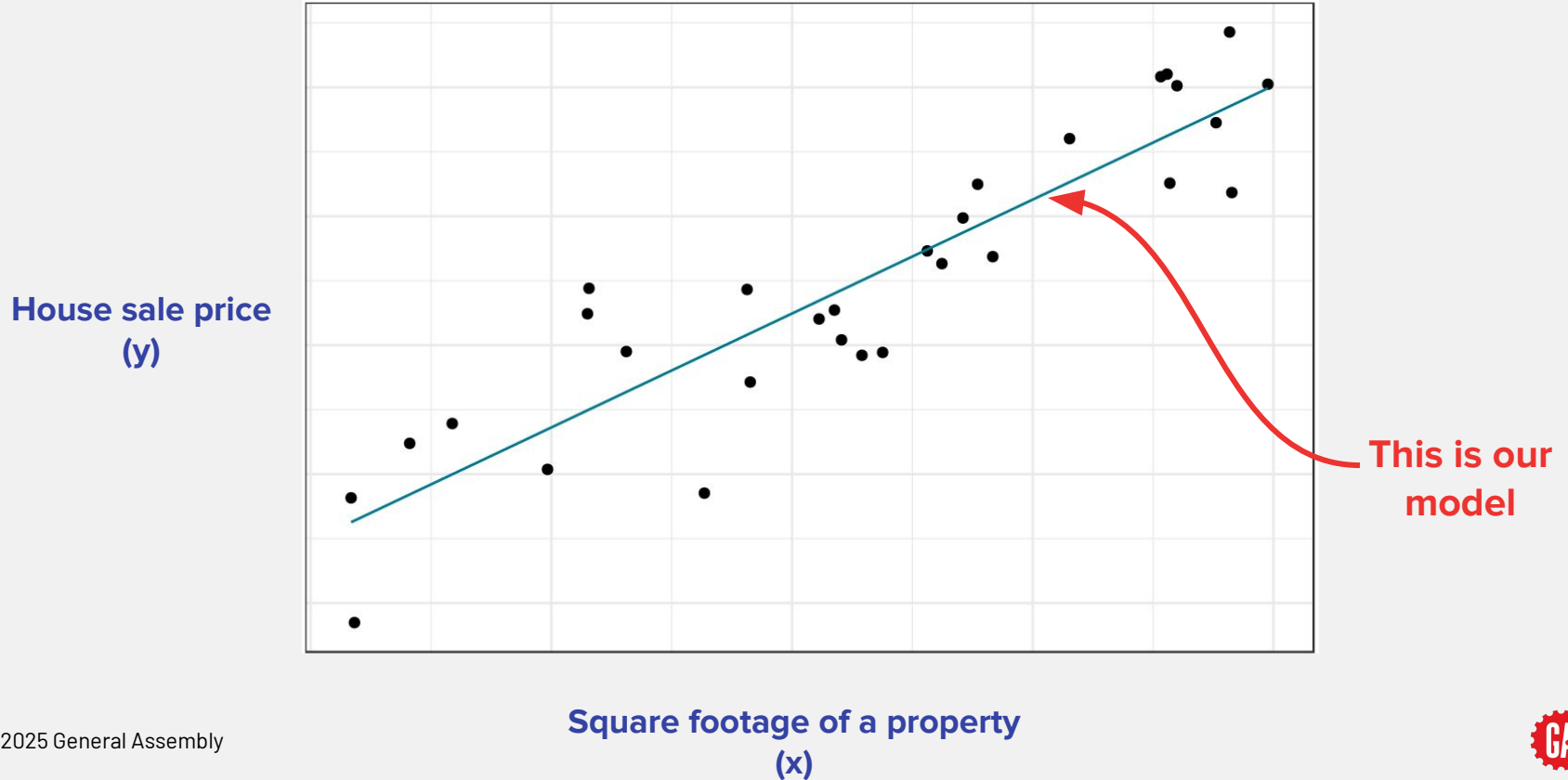


Square footage of a property

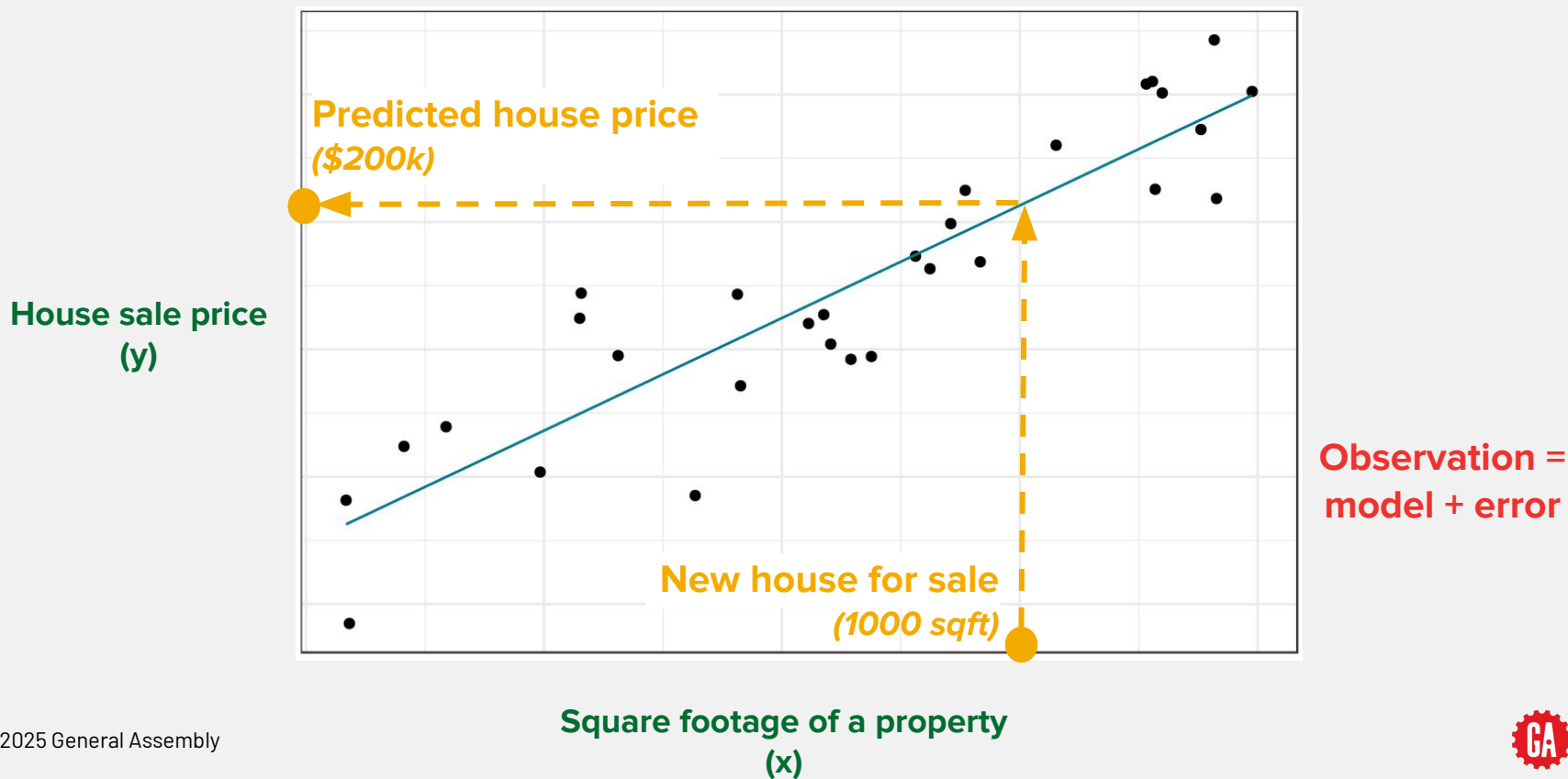
There seems to be a linear relationship between the two variables. We can draw a line to represent that relationship.



There seems to be a linear relationship between the two variables. We can draw a line to represent that relationship.



You can use the model to make a prediction for a given value of x



Let's turn that into math, so we can use it repeatedly

In ordinary least squares linear regression (often just referred to as **OLS**), we try to predict some response variable (y) from at least one independent variable (x). We believe there is a **linear** relationship between the two:

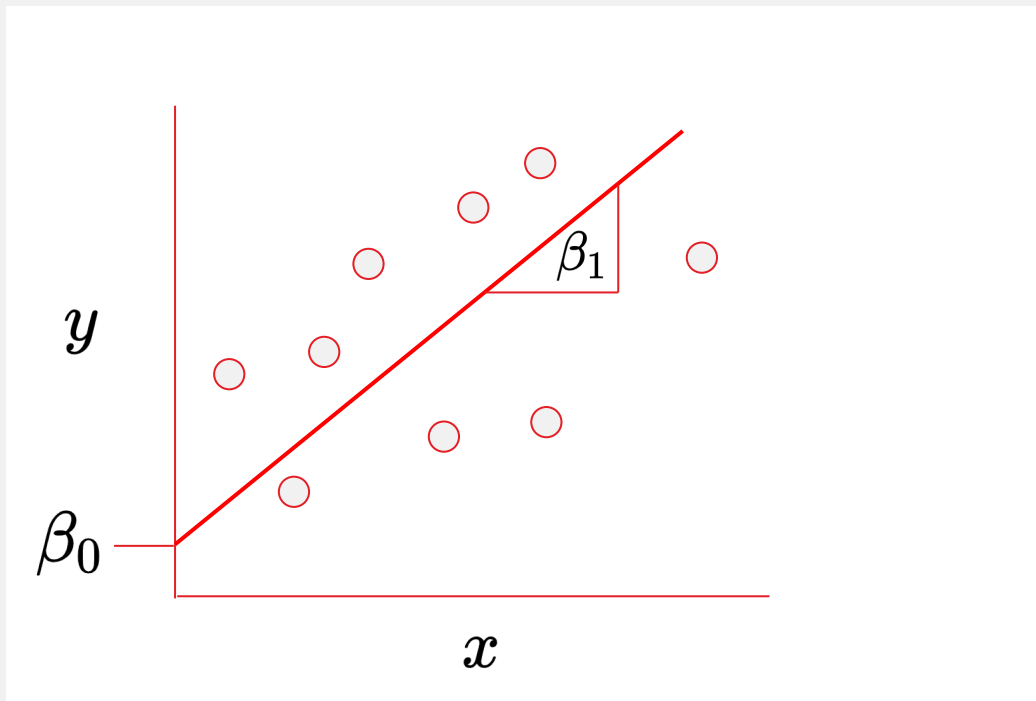
$$y = \beta_0 + \beta_1 x + \varepsilon$$

Where that funny looking “e” stands for “error” - it's random noise inherent in our prediction because nothing will be perfect.

This is just when our model has one independent variable, and is known as Simple Linear Regression.

We'll revisit this later when we want to add more features.

What does the Linear Regression equation mean?



Target
(house price)

Feature
(square feet)

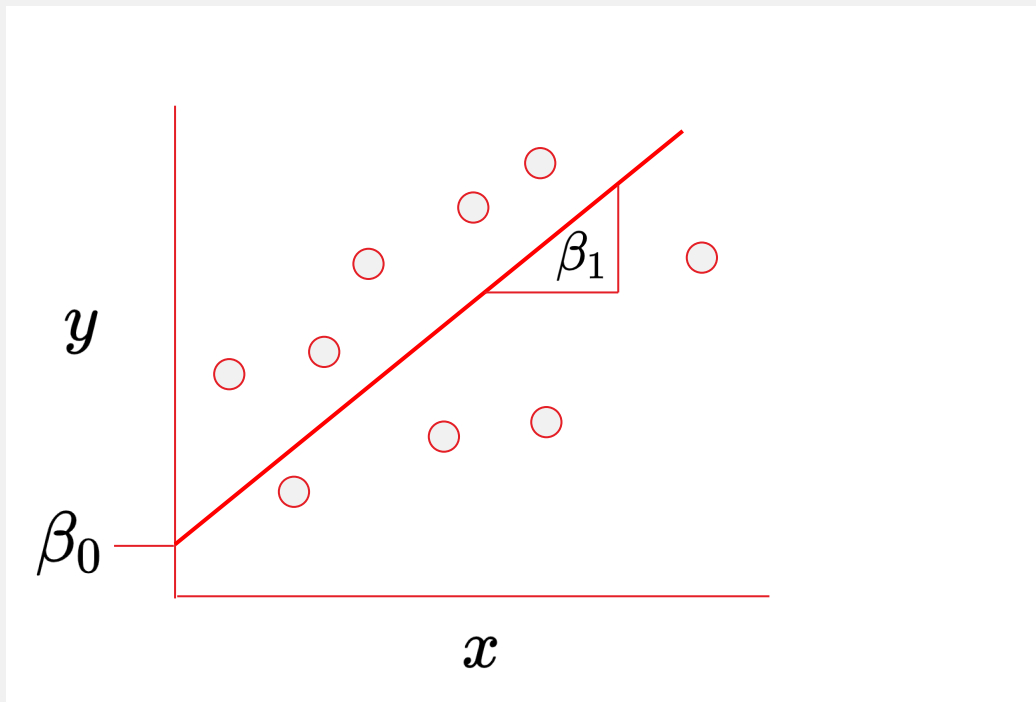
$$\underline{y} = \beta_0 + \beta_1 \underline{x}$$

These are what we are
trying to work out (model):
the coefficients

β_0 (Beta zero)

β_1 (Beta 1)

What does the Linear Regression equation mean?



Target
(house price)

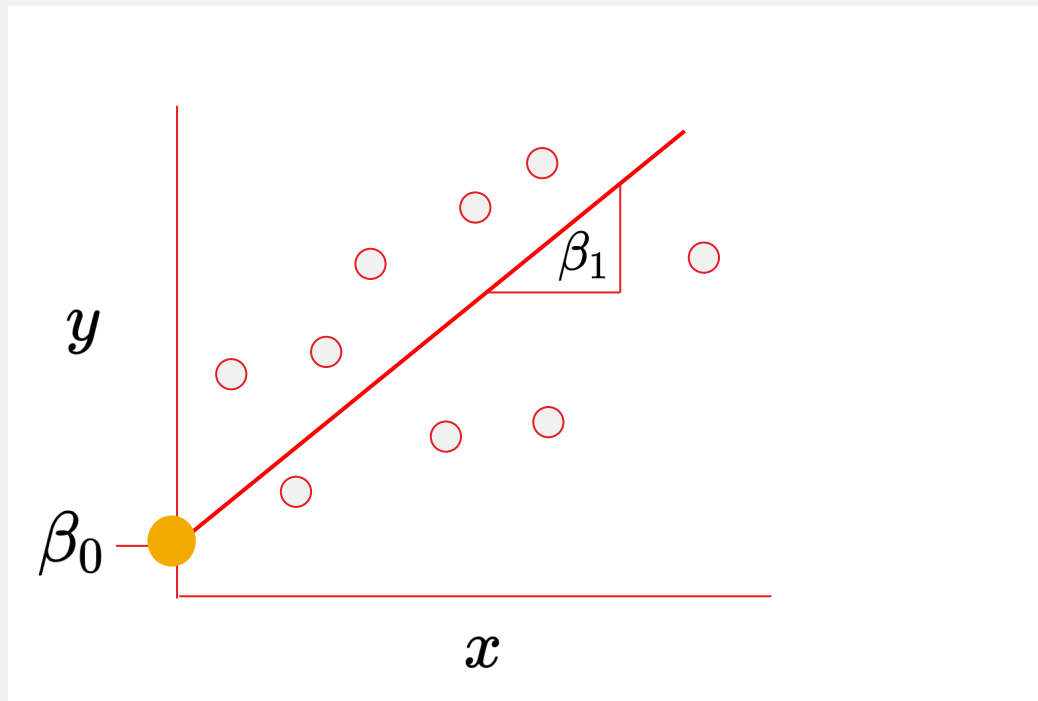
Feature
(square feet)

$$\underline{y} = \beta_0 + \beta_1 \underline{x}$$

These are what we are trying to work out (model):
the **coefficients**

$$y = \beta_0 + \beta_1 x$$

What is β_0 ?



β_0 (Beta zero) is the y-intercept, the expected value of y when x is zero.

Example

$$y = 12000 + 250_1 x$$

Discuss: In your own words, interpret the y-intercept here, where:

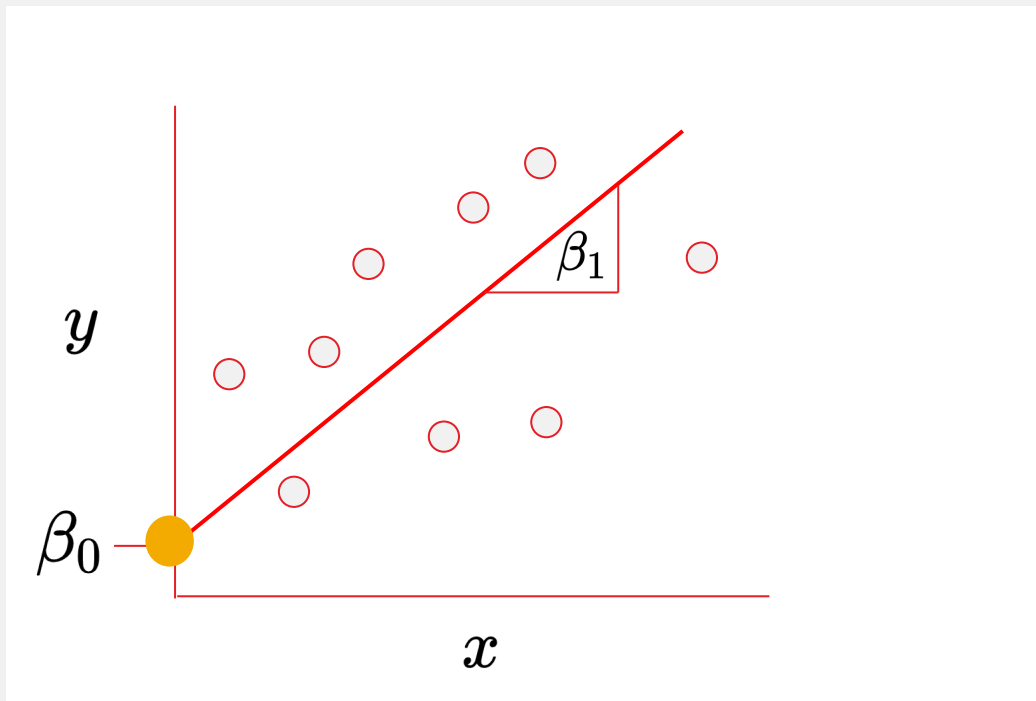
y = house price and

x = square footage.

Does it make sense?

$$y = \beta_0 + \beta_1 x$$

What is β_0 ?



β_0 (Beta zero) is the y-intercept, the expected value of y when x is zero.

Example

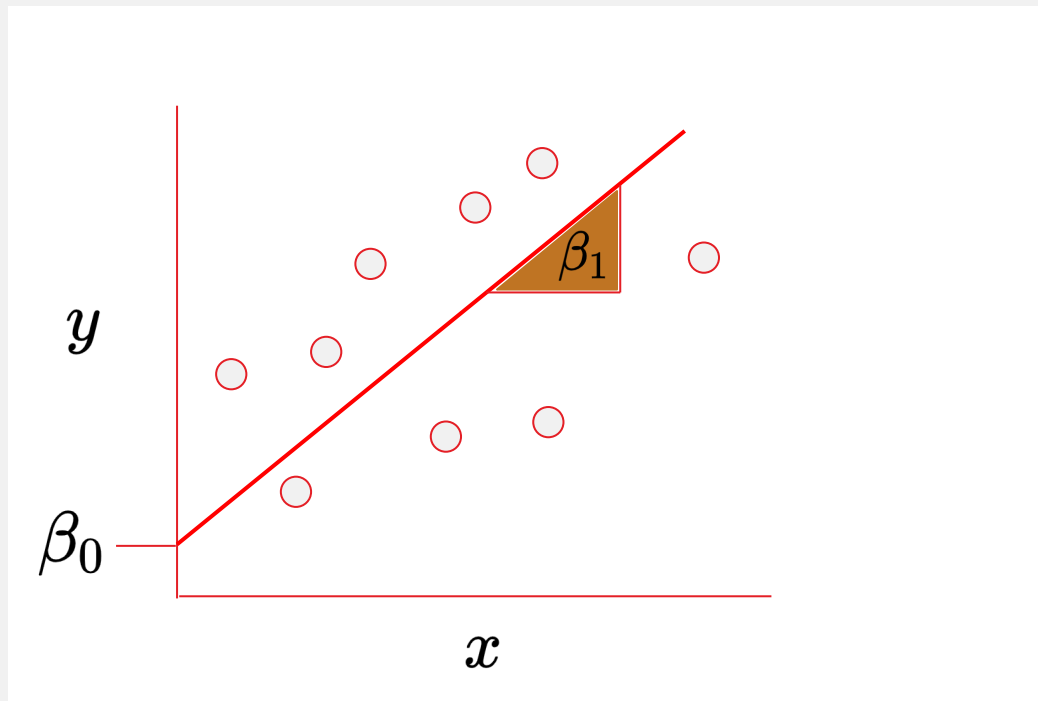
$$y = 12000 + 250_1 x$$

“For a property with no square footage, we expect the sale price to be \$12,000.”

This doesn't make sense. The y-intercept does not always have a sensible interpretation.

$$y = \beta_0 + \beta_1 x$$

What is β_1 ?



β_1 (Beta 1) is the slope: the rate of change between x and y .

Example

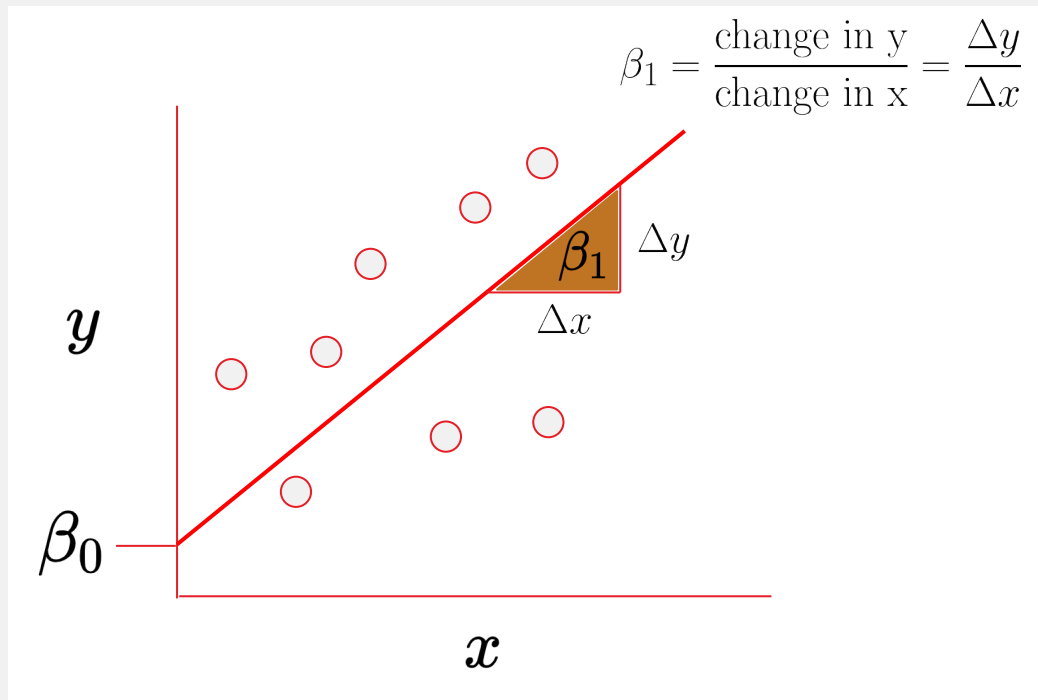
$$y = 12000 + 250_1 x$$

What does the value 250 means here?

It's not intuitive.

$$y = \beta_0 + \beta_1 x$$

What is β_1 ?



β_1 (Beta 1) is the slope: the rate of change between x and y .

Example

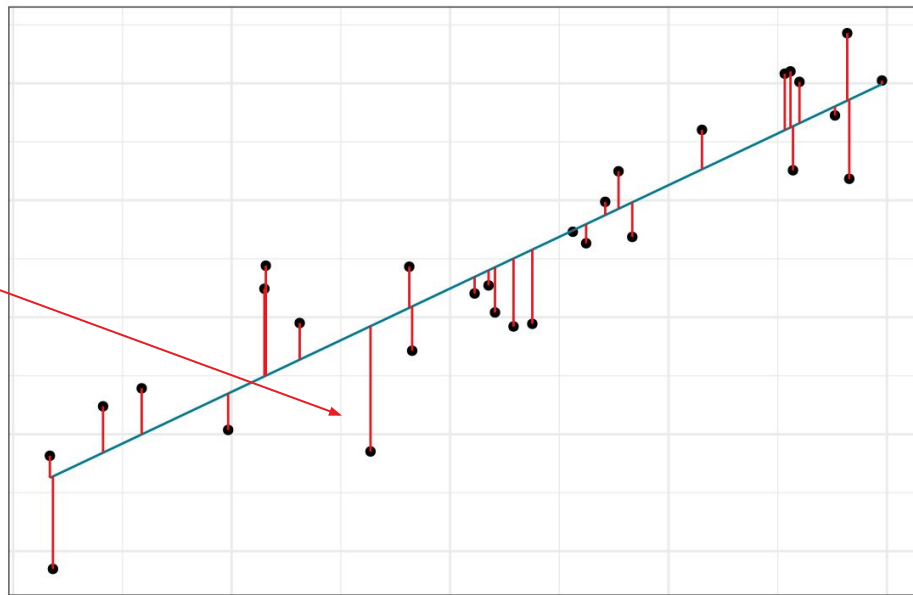
$$y = 12000 + 250_1 x$$

For every square foot that is added to a property, we would expect the sale price of that property to increase by about \$250.

How do we know what the best line is?

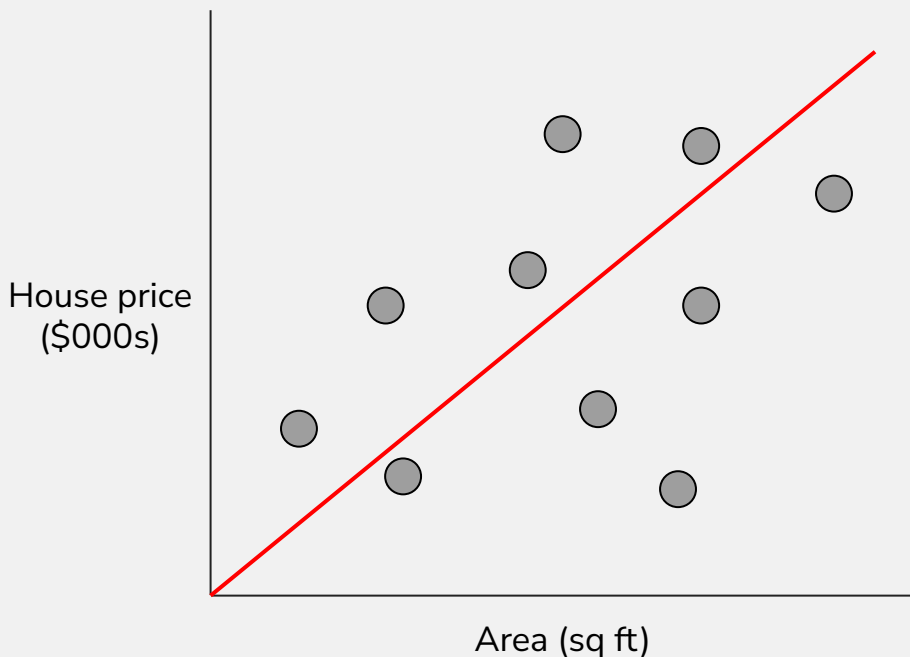
The difference between the actual and the predicted is called a **residual**, and the line of “best fit” minimizes all of these residuals.

Specifically, we minimize the **sum of the squared residuals**, hence the term “least squares”.



$$\text{Observation} = \text{model} + \text{error}$$

Summary: Linear Regression (and Supervised Learning)



Problem: Predict house price using internal area.

This is a **supervised learning** task because it fits the pattern of 'predict **y** using **x**'.

We have **training data** (●) which gives us paired values of house price and area for ten different houses.

We **fit a model** to this data. In this case, it's a very simple straight line (**linear**) model.

We can use the **trained model** (the red line) to predict the price of houses that **aren't** in our training data set, as long as we know their area.

Simple Linear Regression

Let's see it for ourselves!

- Open **02-simple-linear-regression.ipynb**



How Good *is* a Model?

Evaluating a linear regression model

First, some Notation

A boldface $\mathbf{y}$ denotes *all* of our response data. It's **bold** because it's a **vector**.

We typically reserve the letter n to be our **sample size**.

$$\mathbf{y} = (y_1, y_2, \dots, y_n)$$

Notation

We put “hats” over variables to denote they are **predicted values**. That is, “y-hat” represents the predictions based on our original data.

$$\hat{\mathbf{y}} = (\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$$

Notation

Similarly, the betas are things we **estimate**, so they get hats too! Our y-hats are the results of using these estimated values to get predictions.

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

The Residual

Next, we define a **residual** as:

$$e_i = y_i - \hat{y}_i$$

This measures how “off” our predictions were. It’s either positive or negative depending on whether we overestimate or underestimate.

The Sum Squared Error

To get an aggregate measurement of the quality of our model, we often look at the **sum squared error**, or **SSE**:

$$SSE = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$$

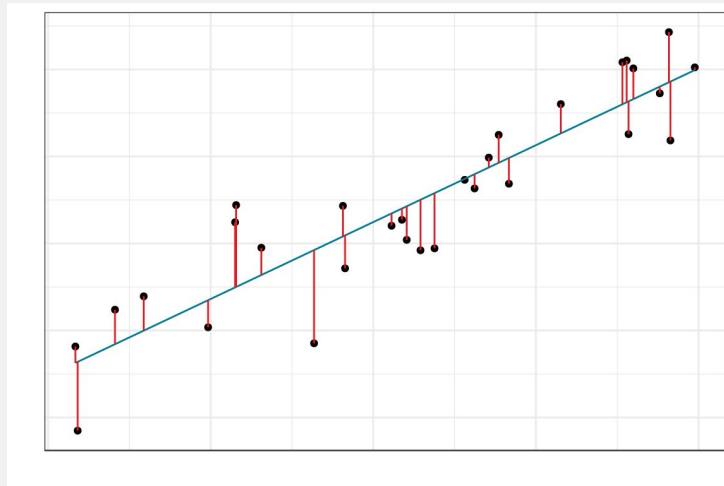
Or more commonly, the **mean squared error (MSE)**:

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum e_i^2$$

Fitting OLS models

Remember, this is the quantity we actually seek to **minimize** in order to find the best values of our betas!

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2 = \frac{1}{n} \sum (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

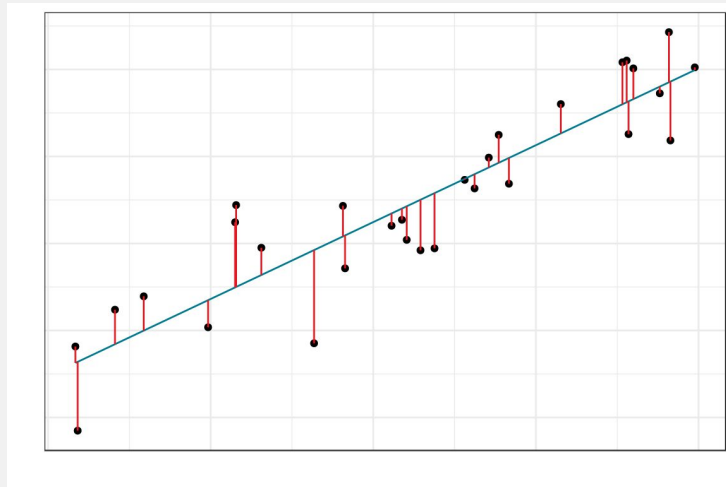


Fitting OLS models

You can also look at the square root of the MSE to see how good a model is.

The advantage of using the root mean squared error (**RMSE**) is that it is in the **same units** as the target variable.

So an RMSE of 2000 if you are trying to predict a house price could be interpreted as '**on average, my model will be wrong by +/- \$2,000 every time it makes a prediction.**'



Another measure that is used is R^2

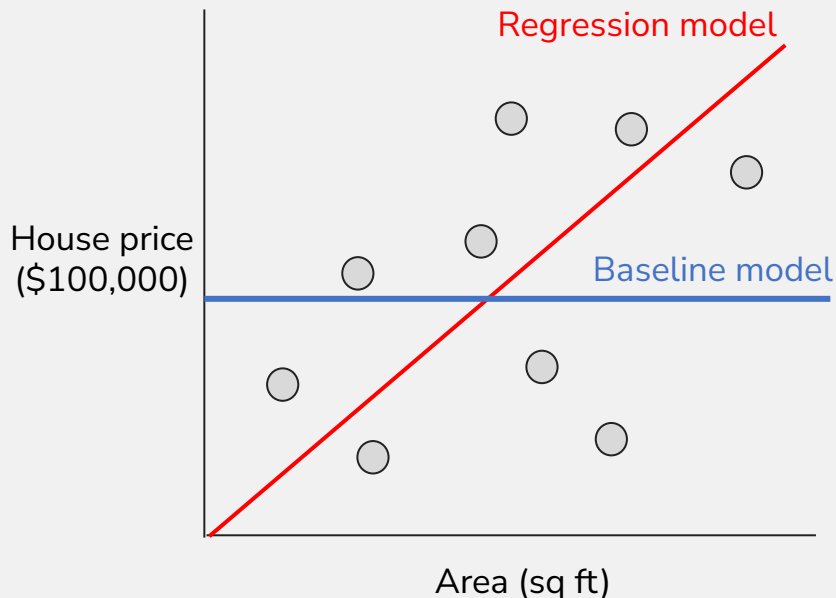
Another popular regression metric is the R^2 , or the **coefficient of determination**.

It is the proportion of variability in y we can explain with x , in other words the proportion of variation your model explains.

Possible values vary from 0 to 1: if R^2 is 1, that means all of the variation is accounted for by the model.

$$R^2 = 1 - \frac{\text{SSE}}{\text{Null SSE}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Baseline Models



A baseline model is a very simple model that always predicts the **same value** of house price, no matter what the input is!

We can calculate the RMSE of the baseline model on the testing data, and use this as a comparison.

If our regression model performance isn't much better than the baseline model, it's not a good sign.

The baseline model is also known as the **null model**.

Simple Linear Regression

Let's check out the results!

- Open **02-simple-linear-regression.ipynb**



Assumptions of Linear Regression

How to check them, and what to do if those assumptions aren't met

OLS Assumptions

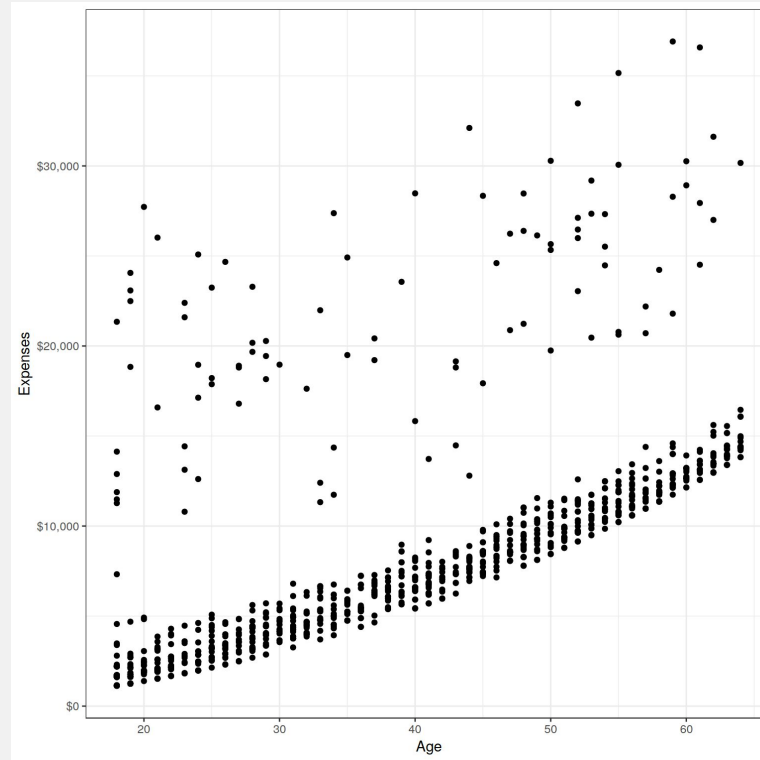
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OLS Assumptions

Conducting OLS comes with some pretty steep assumptions that should be satisfied before believing the results. Luckily, there's a nice acronym to remember them:

- ❑ **L** - Linearity. Relationship between x and y should be approximately linear.
- ❑ **I** - Independence. Your observations should not affect one another.
- ❑ **N** - Normality. Our residuals should be approximately normally distributed.
- ❑ **E** - Equal variances, aka “**homoscedasticity**”. Residuals should have approximately equal variances for each x .

L is for Linearity



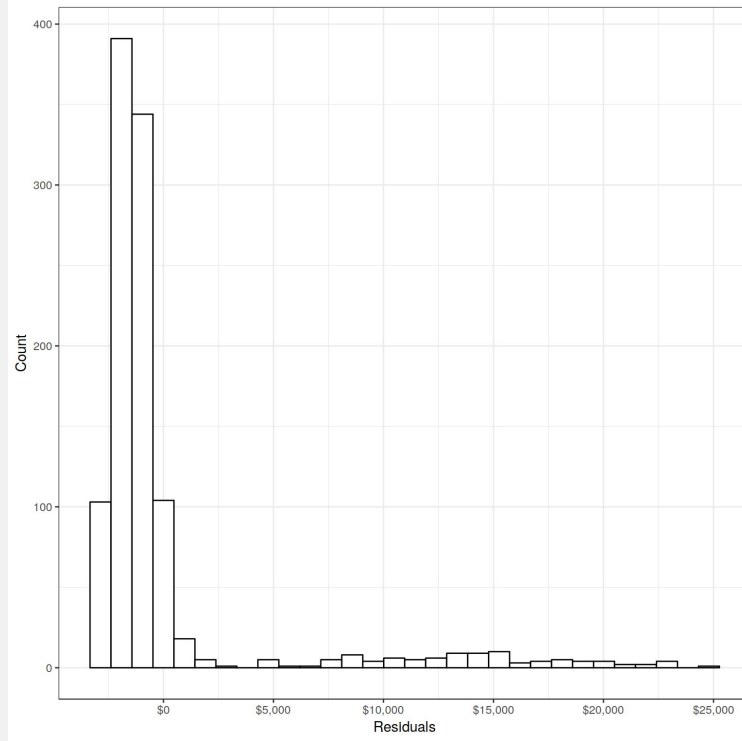
I is for Independence

Are our samples independent from one another?

Yes, these samples were collected independently.

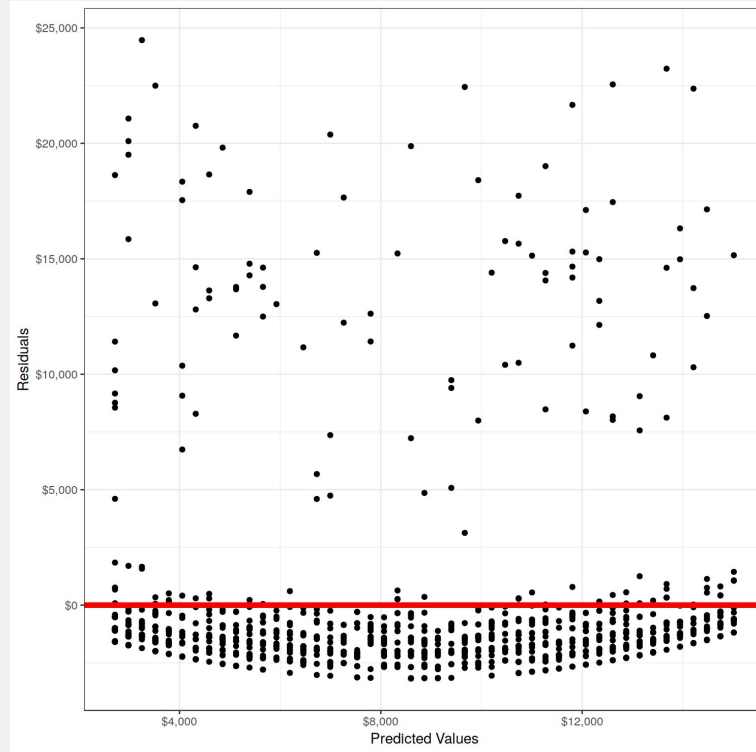
The most common time we'd have to worry about this assumption is when we have **time series data**, when multiple measurements are made on a subject over time.

N is for Normality



This is not normally distributed.

E is for Equal Variances

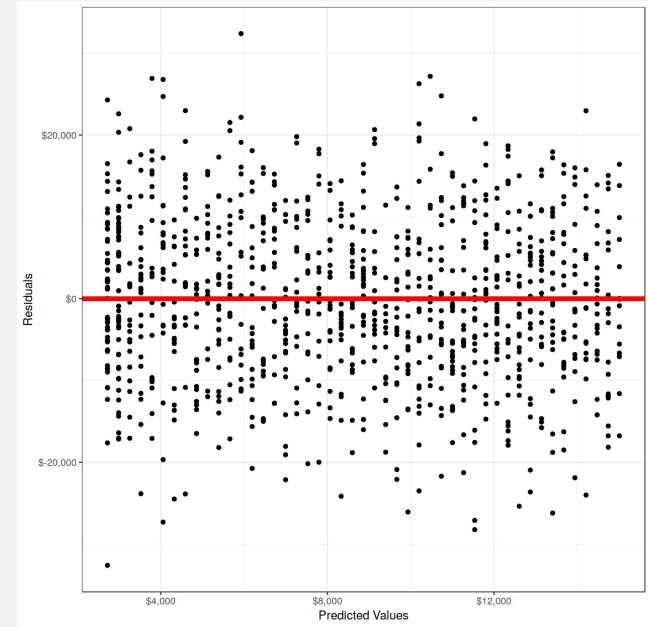


This does not have equal variances for predicted values.

E is for Equal Variance

We want to see **absolute randomness** in our residual plots. That is, no pattern whatsoever. We can clearly see a parabolic pattern in the previous residual plot.

Here's an example of an ideal residual plot.



What to do if our LINE assumptions are violated?!

A common scenario in linear model is when you have:

- A slightly **curvilinear** relationship between x and y
- Very right-skew residuals
- Residuals that tend to spread out from right to left (a “fan shape”)

One quick fix that should improve all of these issues is doing **log regression**. That is, simply take the natural log of y before modeling!

Simple Linear Regression

Let's see if our model passes!

- Open **02-simple-linear-regression.ipynb**

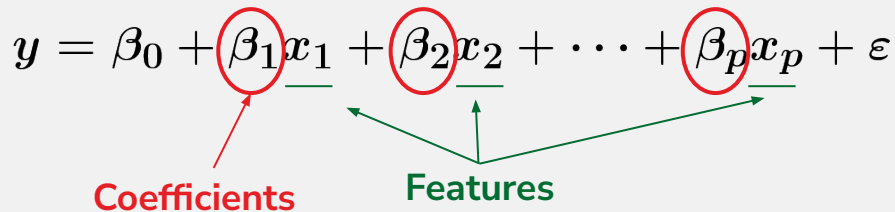


Multiple Linear Regression

An overview

Multiple linear regression is just an extension of simple

When you have more than 1 predictor variable you are doing multiple linear regression.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon$$


The diagram shows the equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon$. The coefficients $\beta_1, \beta_2, \dots, \beta_p$ are circled in red. A red arrow points from the word "Coefficients" to the circle around β_1 . The features $x_1, x_2, \dots, x_p$ are underlined in green. A green arrow points from the word "Features" to the underlines under $x_1, x_2, \dots, x_p$.

There is one more **LINE** assumption to add for multiple regression:

M - No Multicollinearity /
Independence of Predictors

For example, when looking at features for our house pricing model, we may see that the number of cars in the garage is highly correlated with the size of the garage. In this case, we would choose just one of these features to include in our model.

Multiple Linear Regression

Let's see if our model passes!

- Open **03-multiple-linear-regression.ipynb**



Categorical features in models

**One hot encoding
/ dummification**

Categorical Features

How do we work with categorical variables in our model? You have already seen that we can simply use **binary categorical features** as 0/1 variables.

But what if our variable has more than two **levels**?

First some more notation!

Notation: Design Matrix

Mathematically speaking, every time we fit a model, we need a data matrix, sometimes called a **design matrix**:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

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Each **row** is an **observation**.

Notation: Design Matrix

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Each **column** is a variable.

Notation: Design Matrix

Mathematically speaking, every time we fit a model, we need a data matrix, sometimes called a **design matrix**:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

The first column is all 1s and corresponds to the **intercept**. (sklearn handles this automatically)

Notation: Matrix Notation of OLS

Why? This allows us to write OLS like this:

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

This is just a way to condense this:

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}_{n \times 1}} = \underbrace{\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}}_{\mathbf{X}_{n \times p+1}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix}}_{\beta_{p+1 \times 1}} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{\varepsilon_{n \times 1}}$$

Dummy Columns

For a categorical variable with k levels, we need to make one **dummy column** for each:

Season
Summer
Spring
Spring
Fall
Fall
Summer
Winter



$X =$

	W	Sp	Su	F
1	0	0	1	0
1	0	1	0	0
1	0	1	0	0
1	0	0	0	1
1	0	0	0	1
1	0	0	1	0
1	1	0	0	0

Dummy Columns

But wait! This is actually not ok - the intercept term is simply the sum of all of these four columns! In linear algebra terms, this is called being **rank-deficient**, and will make our model impossible to fit.

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Dummy Columns

So, we next have to **drop any column**.

The default in pandas is to drop the first. This means our first column (in this case, Winter) corresponds to the **baseline category**. When interpreting, everything is **relative to this column**.

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Dummy Columns

So, we next have to **drop any column**.

The default in pandas is to drop the first. This means our first column (in this case, Winter) corresponds to the **baseline category**. When interpreting, everything is **relative to this column**.

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

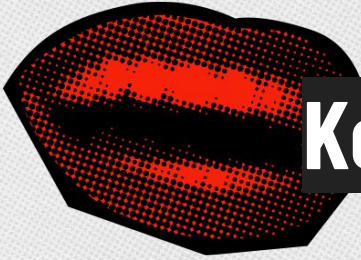
Multiple Linear Regression

Now you try.

- ☐ Open **02-linear-regression.ipynb**



Bring It Home



Key takeaways

Linear Regression



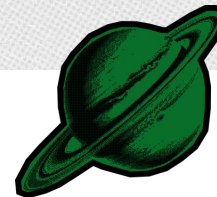
Strengths

Very easy to interpret, and the impact of each variable on the target is quantified (the coefficients are interpretable)



Weaknesses

Lots of assumptions need to be met (LINEM), in particular the linear relationship between variables and the target



Metrics

Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error, R^2

The core steps to take for Linear Regression

① Assemble predictors (X) and target (y)

② Instantiate the model

③ Fit the model

⑥ Evaluate the model

④ Check out and interpret the coefficient(s)

⑤ Make predictions

```
from sklearn.linear_model import LinearRegression
X = (must be df)
y = (must be Series)
linreg = LinearRegression()
```

```
linreg.fit(X_train, y_train)
```

```
R2 = linreg.score
```

OR
from sklearn import metrics

```
RMSE = np.sqrt(metrics.mean_squared_error(y,  
linreg.predict(X))
```

RMSE vs baseline (RMSE of mean)

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \varepsilon$$

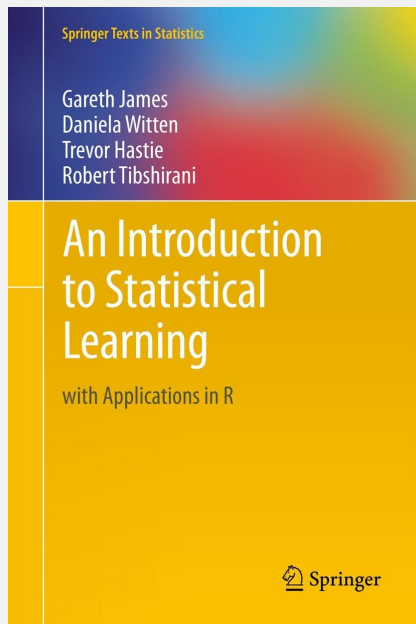
```
linreg.predict()
```

FAQ / common pitfall

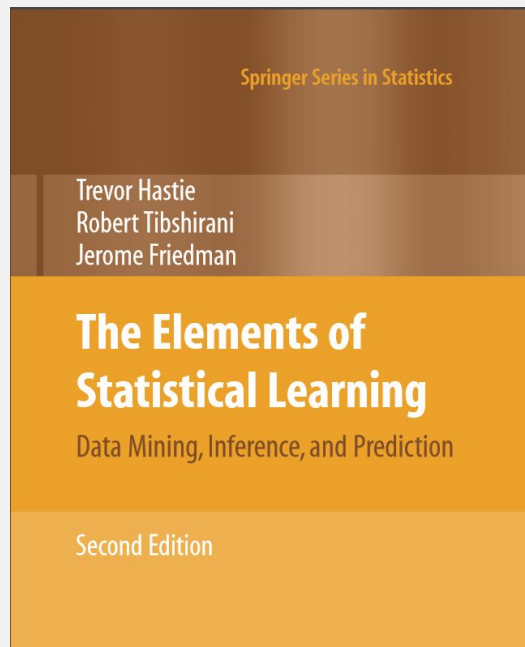
Do the coefficients show which variables have the biggest impact on the target?

No they don't. They indicate the relationship between the variable and the target, all else being equal. So they reflect the impact of each variable in its own units. You can only compare across all variables if they are all in the same unit.

The Data Science Book Club



Undergraduate math level, very readable



All topics ISL has but at the graduate math level. A few additional chapters.

